

# Meridian Logistics — System Architecture

## Services

The Meridian platform is built from four core services. The Shipment Service is the system of record for every parcel and calls the Routing Service to compute delivery routes. The Routing Service uses the Mapbox Directions API for road geometry. The Tracking Service ingests scan events from handheld devices and exposes a public tracking API. The Fleet Service manages the roster of drivers and vehicles.

## Data stores

The Shipment Service persists parcel data in a PostgreSQL database named `shipments_db`. The Tracking Service writes scan events to a ClickHouse cluster for fast time-series queries. All services publish domain events to a RabbitMQ broker on the exchange `'meridian.events'`. The Routing Service caches computed routes in Redis for 6 hours.

## Authentication

Internal services authenticate with mutual TLS. External API clients use API keys issued by the Gateway Service; each key is scoped to a single customer account and rate-limited to 600 requests per minute. Keys can be rotated from the customer portal and old keys stay valid for a 24-hour grace period.

## Deployment

Meridian runs on Google Kubernetes Engine in the `eu-west-1` region. Each service runs with a minimum of 2 replicas and autoscales on CPU. Images are built by CircleCI and stored in Google Artifact Registry. Production releases happen on Tuesdays and Thursdays and require sign-off from the Infrastructure team.